

Introduction

Handle decompositions provide a systematic way to build compact manifolds by attaching elementary handles indexed by dimension. These decompositions arise naturally from Morse theory. For 3-manifolds this leads to Heegaard splittings and Heegaard diagrams. For 4-manifolds, the analogous construction yields Kirby diagrams. This is where knot theory enters fundamentally, 2-handles correspond to framed knots and links, directly shaping the topology of the 4-manifold.

CW-decomposition via Morse Lemma

Definition (Morse Function): A smooth map $f : M \rightarrow \mathbb{R}$ on a manifold M is a Morse function if all of its critical points are **non-degenerate**, that is, its Jacobian A is non-singular, $\det(A) \neq 0$.

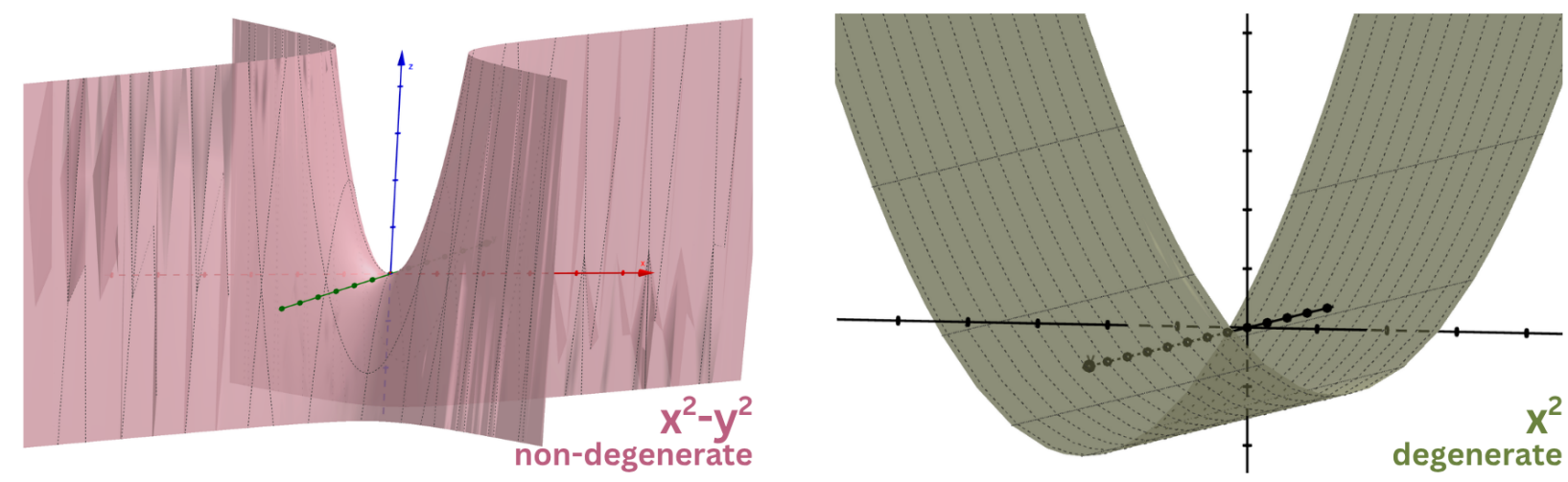


Figure 1. Examples of degenerate and non-degenerate critical points.

Theorem (Morse Lemma): Let f be a smooth function on M and let p be a non-degenerate critical point of f . There exists a local coordinate system $(x^1, \dots, x^n)$ in a neighborhood U of p s.t.

$$f = f(p) - (x^1)^2 - \dots - (x^\lambda)^2 + (x^{\lambda+1})^2 + \dots + (x^n)^2$$

holds throughout U , where λ is the index of f at p , and $x^i(p) = 0$.

- For M smooth and compact, any continuous function can be “approximated” by a Morse function, that is, Morse functions are **generic** in C^2 topology.
- For M smooth, closed, and connected, M admits a **self-indexing** Morse function (i.e. $f(c) = i(c)$) with one critical point of index 0 and index n each.
- The topology of sublevel sets of f changes in one of two ways (Fig 2).

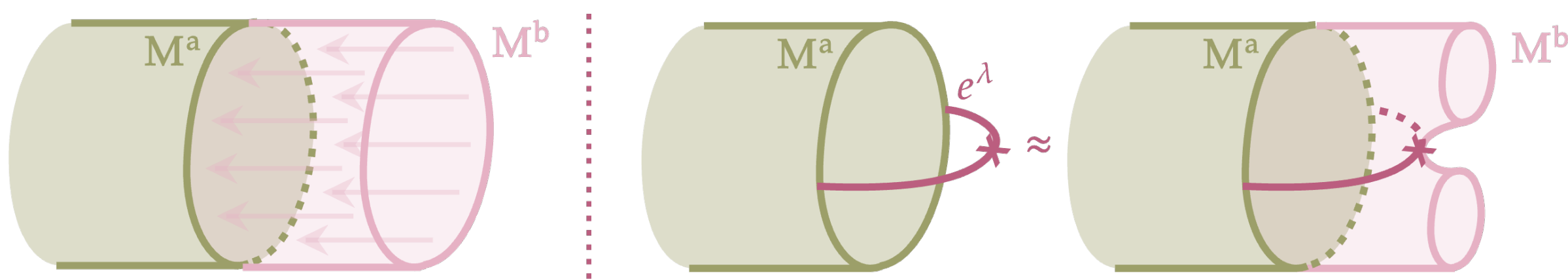


Figure 2. Between sublevel sets $M^a = f^{-1}([-\infty, a])$ and $M^b = f^{-1}([-\infty, b])$ for $a < b$, two cases arise. Case 1 (Left): If there are no critical points between a and b , then M^a is a deformation retract of M^b thus $M^a \simeq M^b$. Case 2 (Right): if there is a critical point of index λ , then $M^a \cup e^\lambda$ is a deformation retract of M^b , hence $M^a \cup e^\lambda \simeq M^b$.

- This gives rise to a **CW-decomposition** of a compact manifold, which has an analogous construction for **handles**. Further details can be found in [Mil63].

Theorem (CW-decomposition): Let $f : M \rightarrow \mathbb{R}$ be a Morse function on a compact manifold M . Then M admits a CW-decomposition with exactly one λ -cell for each critical point of index λ .

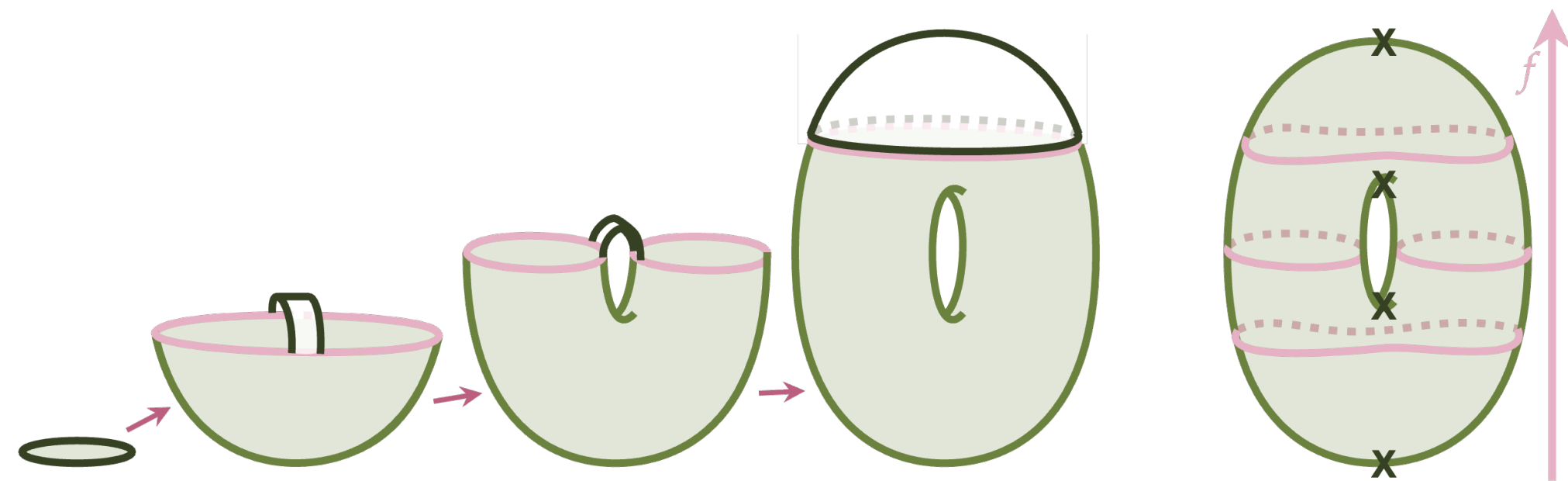


Figure 3. CW-decomposition of a torus via a height function as the Morse function. Each stage corresponds to passing through a critical point: attaching a 0-cell (minimum), two 1-cells (saddle points), and a 2-cell (maximum).

Theory of Handle Decompositions

Definition (Handle): Let M be a smooth n -manifold. For $0 \leq k \leq n$, an n -dim k -handle H is a copy of $D^k \times D^{n-k}$ attached to ∂M along $\partial D^k \times D^{n-k}$, via a smooth map $\phi : \partial D^k \times D^{n-k} \rightarrow \partial M$.

Theorem (Handlebody decomposition of a compact manifold M): Let $f : M \rightarrow \mathbb{R}$ be a Morse function on a compact manifold M . Then M admits a handle decomposition with exactly one λ -handle for each critical point of index λ .

Handlebody Decomposition for 3-Manifolds: Heegaard Diagrams

- Choose a self-indexing Morse function on a closed oriented 3-manifold M with one critical point of index 0 and 3 each.
- We obtain the **Heegaard splitting**, $M = H_1 \cup_{\Sigma_g} H_2$, where $H_1 = M^{[0, \frac{2}{3}]}$, $H_2 = M^{[\frac{2}{3}, 3]}$, and $\partial H_1 \simeq \partial H_2 \simeq \Sigma_g$ a closed, orientable, connected surface of genus g .
- Encode the gluing information of 2-handles with **Heegaard diagrams** (Fig 4).

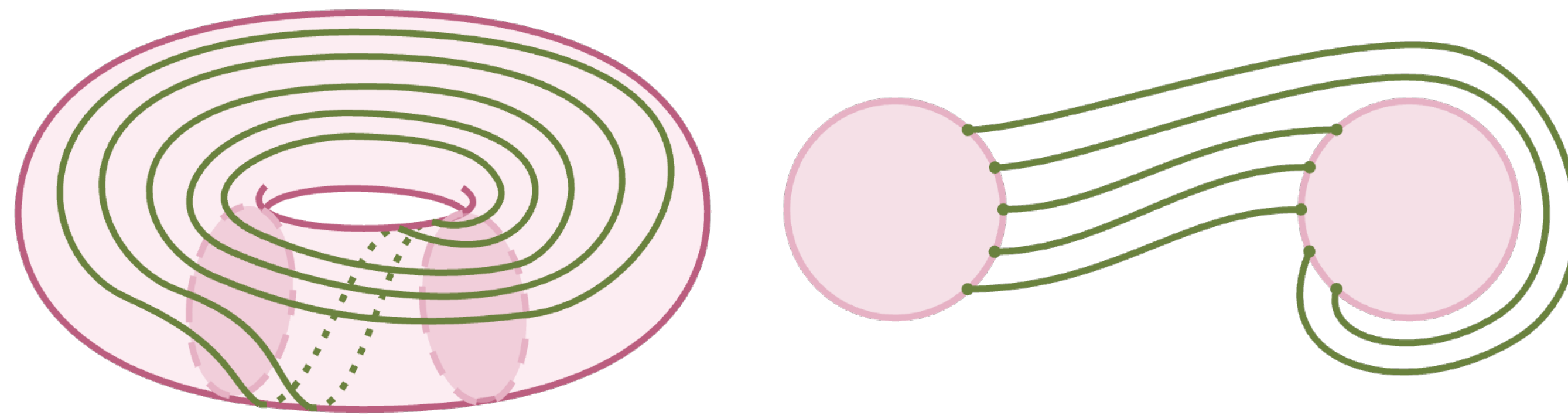


Figure 4. Heegaard diagram for Lens Space $L(5, 2)$. **Left:** The solid torus is the union of the 0- and 1-handle, the green closed loop is the attaching sphere of the 2-handle. **Right:** Alternative representation in $\mathbb{R}^2$. Puncture the 0-handle boundary to obtain $\mathbb{R}^2$, represent 1-handles as pairs of discs, and attaching spheres of 2-handles as arcs that become circles under identification of disc boundaries by reflection.

Handlebody Decomposition for 4-Manifolds: Kirby Diagrams

- Choose a self-indexing Morse function on a closed oriented 4-manifold M with one critical point of index 0 and 4 each.
- The construction of the **Kirby diagram** of M is analogous to the $\mathbb{R}^2$ Heegaard diagram.
- Attaching regions of 1-handles are pairs of balls. Attaching regions of 2-handles are represented with arcs that form framed links (Fig 5). Framing is discussed in the next section.
- Attachment of 3- and 4-handles is unique for closed manifolds. More details in [GS99].

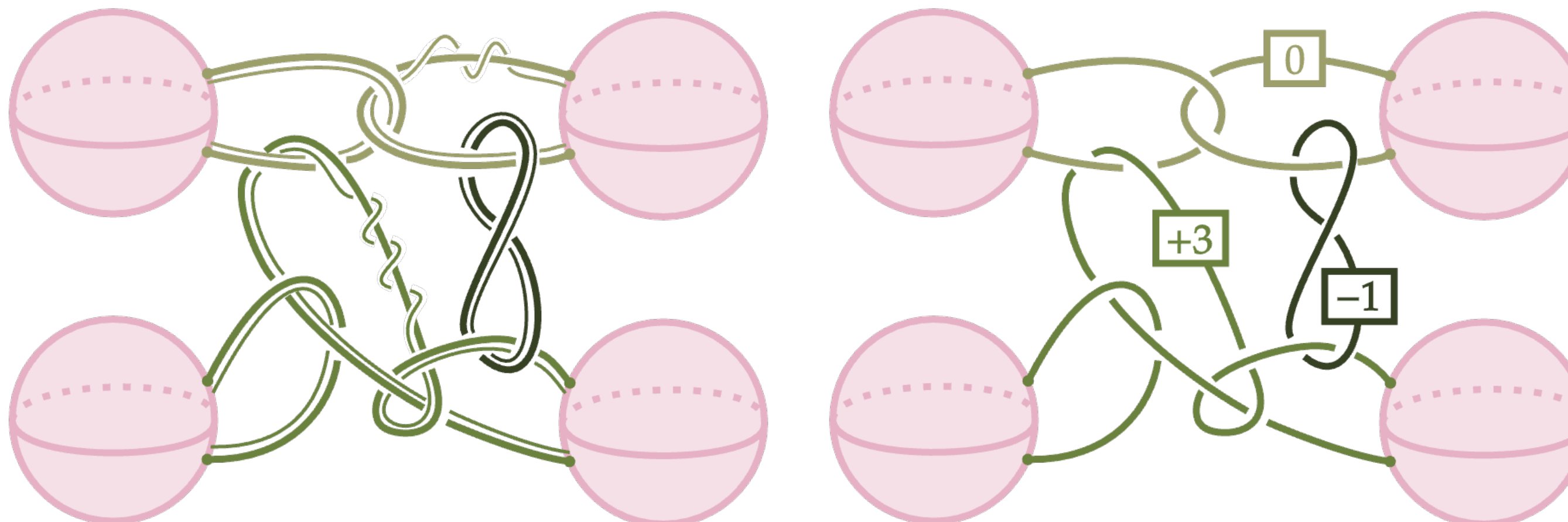


Figure 5. Kirby diagram for a 4-manifold. **Left:** Attaching regions of 1- and 2-handles as framed links in $\mathbb{R}^3$. **Right:** The equivalent diagrammatic simplification, where geometric twists are replaced by explicit integer framing coefficients encoding the $\mathbb{Z}$ -framing of each 2-handle.

Enter Knot Theory

The attachment of a k -handle, $\phi : \partial D^k \times D^{n-k} \rightarrow \partial X$, is determined by the **embedding** $\phi_0 : \partial D^k \times 0 \rightarrow \partial X$, and a **framing** of $\phi_0(\partial D^k)$ (an identification of the normal bundle of $\phi_0(\partial D^k)$ with $S^{k-1} \times \mathbb{R}^{n-k}$).

- Unique handle framings up to isotopy correspond bijectively to elements of the homotopy group $\pi_{k-1}(SO(n-k))$. Trivial for 3D 1-, 2-handles and 4D 1-, 3-handles, thus attachments are unique.
- For 4D 2-handles, $\pi_1(SO(2)) \cong \mathbb{Z}$. Framings are encoded strictly by integers representing full ribbon twists on 1-dim knots in the 3-boundary (Fig 5).
- Knots completely encode the structural data of smooth spaces, providing relators for fundamental groups and generating intersection forms of 4-manifolds.
- Allows topologists to classify and study smooth 3- and 4-manifolds purely via diagrammatic manipulation of ribbons and knots.

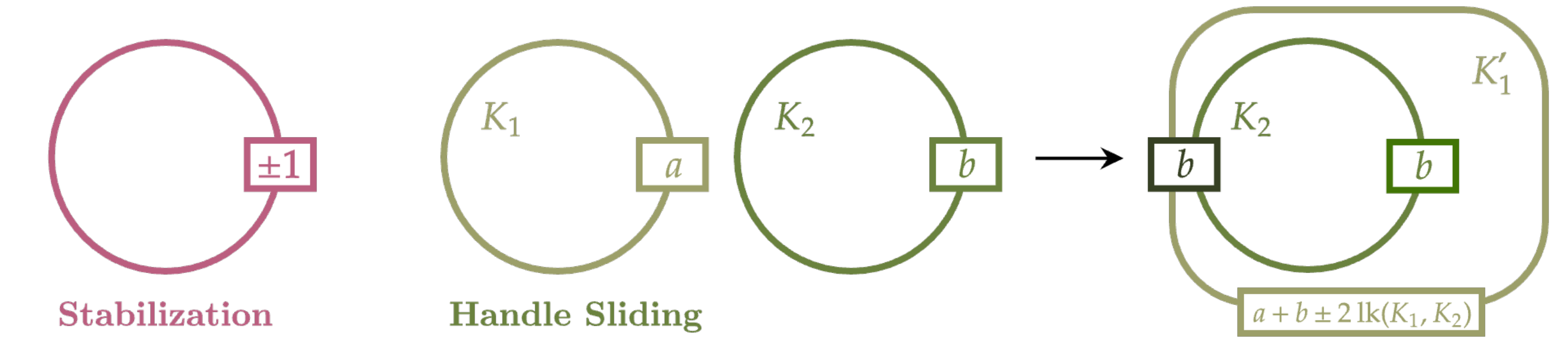


Figure 6. **Left:** Kirby Move 1 (**Stabilization**), insertion or deletion of a ± 1 -framed unknot unlinked from the rest of the diagram, preserving the boundary 3-manifold while changing the 4-manifold interior. **Right:** Kirby Move 2 (**Handle Slide**), sliding a 2-handle K_1 over a parallel ribbon band of K_2 , where $\text{fr}(K'_1) = \text{fr}(K_1) + \text{fr}(K_2) \pm 2 \cdot \text{lk}(K_1, K_2)$.

Theorem: (Kirby’s Theorem) Two framed links in S^3 yield diffeomorphic 3-manifolds if and only if they are related by a finite sequence of **Kirby moves** (Fig 6, [GS99]).

Recent work related to Knot theory and Kirby Calculus

The Conway knot is not slice ([Pic20])

The **Knot Trace** $X(K)$, is the 4-manifold obtained by gluing a 0-framed 2-handle to B^4 along K . Knots with the same knot trace is either both slice or both not slice. Piccirillo constructed a mutant knot K' (Fig 7) and proved $X(K') \cong X(K)$ via Kirby calculus. By showing K' is not slice, she proved the Conway knot is not slice. For an expository article, see [Hom21].

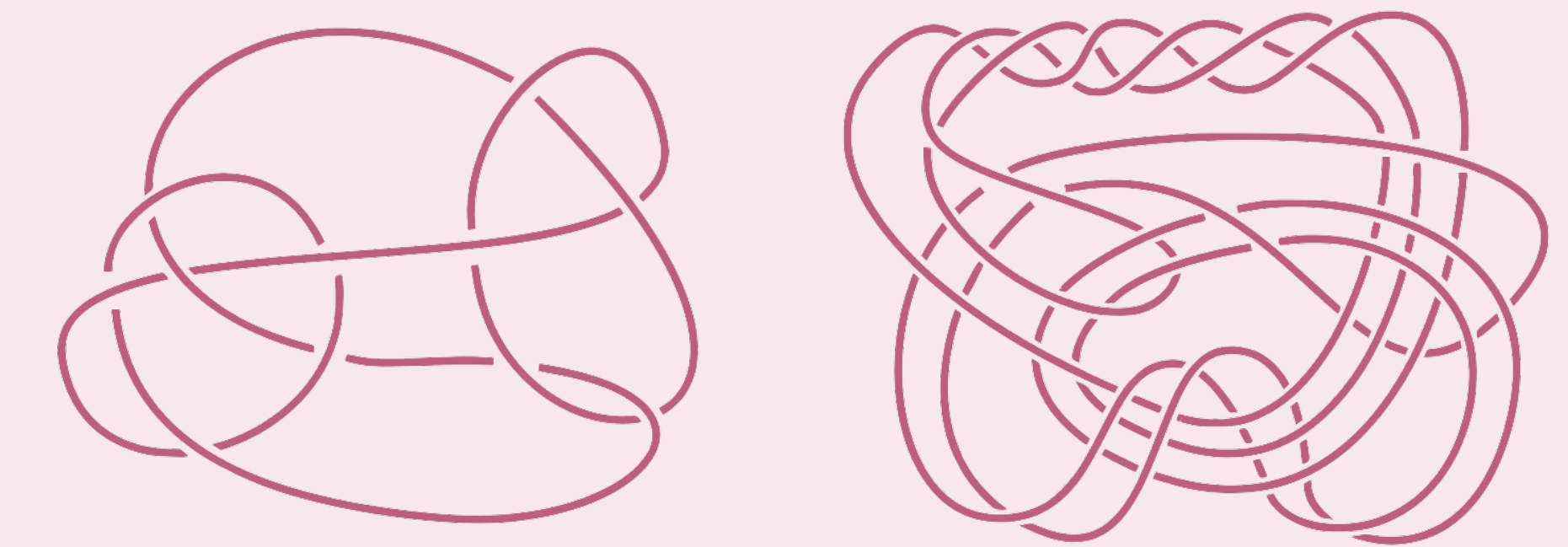


Figure 7. **Left:** Conway knot. **Right:** Piccirillo’s construction. Diagrams based on [Hom21].

Acknowledgements

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References

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